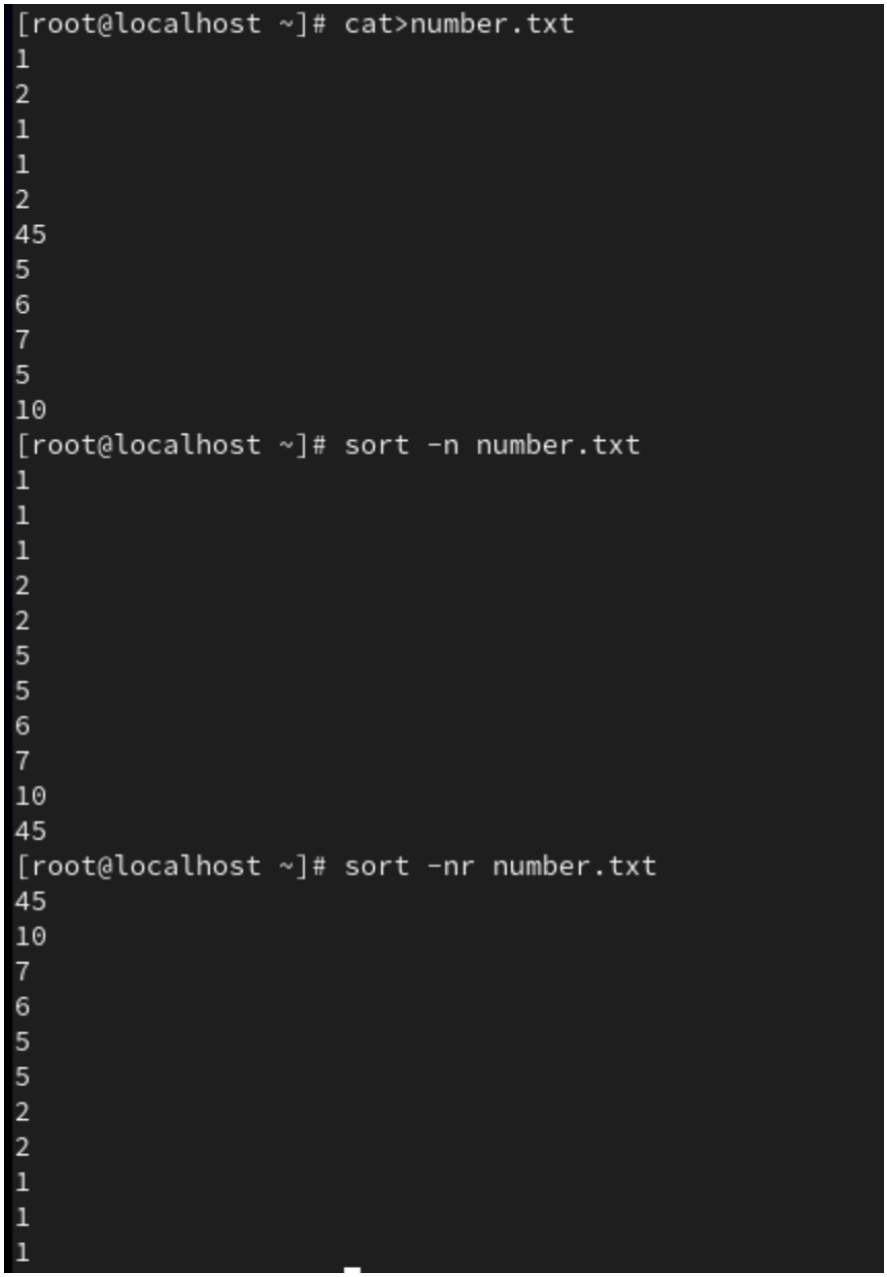
Sort

sort -n filename	for numeric sort
sort -nr filename	for reverse numeric sort
sort -M filename	for Month sort
sort -R filename	for random sort



Sed Editor

```
[root@localhost ~]# cat>colors.txt
hemant jatal
[root@localhost ~]#
[root@localhost ~]# sed -i 's/hemant/pratibha/' colors.txt
[root@localhost ~]# cat colors.txt
pratibha jatal
[root@localhost ~]# sed 's/pratibha/hemant/' colors.txt
hemant jatal
[root@localhost ~]# cat colors.txt
pratibha jatal
```

```
[root@localhost ~]# cat os.txt
cat: os.txt: No such file or directory
[root@localhost ~]# cat>os.txt
Unix is great os. unix is opensource. Unix is free os.
Unix is cli based os. Unix is powerful.
Unix is easy to learn. Unix is a multiuser os. Learn Unix
[root@localhost ~]# sed 's/Unix/linux/' os.txt
linux is great os. unix is opensource. Unix is free os.
linux is cli based os. Unix is powerful.
linux is easy to learn. Unix is a multiuser os. Learn Unix
[root@localhost ~]# sed 's/Unix/linux/2' os.txt
Unix is great os. unix is opensource. linux is free os.
Unix is cli based os. linux is powerful.
Unix is easy to learn. linux is a multiuser os. Learn Unix
[root@localhost ~]# # second occurace in every line
[root@localhost ~]# # replace all occurace in every line
[root@localhost ~]# sed 's/Unix/linux/g' os.txt
linux is great os. unix is opensource. linux is free os.
linux is cli based os. linux is powerful.
linux is easy to learn. linux is a multiuser os. Learn linux
[root@localhost ~]# sed '$d' os.txt # delete the last line
Unix is great os. unix is opensource. Unix is free os.
Unix is cli based os. Unix is powerful.
[root@localhost ~]# cat>data.txt
```

cat -n sample.txt #to view content with line number
#sed -i 's/^$/Linux/g' sample.txt ^$---->blank lines
#cat sample.txt

```
[root@localhost ~]# cat>data.txt
JSmith  10001   IT      Chennai
John    10002   Prod    Hyderabad
Mary    10003   IT      Bangalore
Albert  10004   HR      Pune
Simon   10005   HR      Mumbai
Ricky   10006   SALES   Pune
[root@localhost ~]# #a. Delete lines containing "SALES"
[root@localhost ~]# sed '/SALES/d' data.txt
JSmith  10001   IT      Chennai
John    10002   Prod    Hyderabad
Mary    10003   IT      Bangalore
Albert  10004   HR      Pune
Simon   10005   HR      Mumbai
```

If you want to replace every occurrence in a line, combine it with the global `g` flag:

bash

```
sed 's/apple/orange/gI' filename.txt
```

Encryption and Decryption

```
[root@localhost ~]# gpg -c colors.txt
gpg: directory '/root/.gnupg' created
gpg: keybox '/root/.gnupg/pubring.kbx' created
[root@localhost ~]# ls
anaconda-ks.cfg  colors.txt.gpg  Desktop    Downloads  hemant.txt  number.txt  Pictures  Templates
colors.txt       data.txt        Documents  hello      Music       os.txt      Public    Videos
[root@localhost ~]# rm -rf colors.txt
[root@localhost ~]# gpg colors.txt.gpg
gpg: WARNING: no command supplied.  Trying to guess what you mean ...
gpg: AES256.CFB encrypted data
gpg: encrypted with 1 passphrase
[root@localhost ~]# ls
anaconda-ks.cfg  colors.txt.gpg  Desktop    Downloads  hemant.txt  number.txt  Pictures  Templates
colors.txt       data.txt        Documents  hello      Music       os.txt      Public    Videos
[root@localhost ~]# cat colors.txt
pratibha jatal
```

GREP

Command / Tag	Short note
`grep`	Searches for a pattern in a file.
`-i`	Ignores case sensitivity (Mumbai = MUMBAI).
`-v`	Displays lines that do not match the pattern.
`-c`	Counts the number of matching lines.
`-n`	Displays matching lines with line numbers.
`-o`	Displays only the matched text.
`^`	Matches text at the beginning of a line.
`$`	Matches text at the end of a line.

`grep "Mumbai" city.txt`	Search for exact case-sensitive text
`grep -i "mumbai" city.txt`	Ignore uppercase/lowercase differences
`grep -v "Mumbai" city.txt`	Display lines that do not contain Mumbai
`grep -n "Mumbai" city.txt`	Display matching lines with line numbers
`grep -c "Mumbai" city.txt`	Count matching lines
`grep -l "Mumbai" *.txt`	Display filenames containing Mumbai
`grep -w "Mumbai" city.txt`	Match the whole word
`grep -o "Mumbai" city.txt`	Display only the matched text
`grep -r "Mumbai" /home/`	Search recursively inside files in a directory

Find and locate

```
Find & Locate command in Linux:  used to find/search a folder/files in a Linux system

Nautilus:  is the file manager for the GNOME desktop. It allows one to browse directories, preview files.

find command is used to find files/folders in Linux system.

1. To find file inittab in entire file system
        #find / -name inittab


2. To find file passwd in entire file system(ignore case)
        #find / -iname passwd

3. To find files which are empty in /root directory
        #cd
        ~]#find . -type f  -empty

4. To find files which are empty not in /root directory
        #cd
        ~]#find . -type f  ! -empty                f-->file

5. To find all folders named as London in entire file system
        ~]#find / -type d -iname London                d-->directory


6. To find files whose size is more than +10KB in /root
        #cd
        ~]#find . -size +10k
```

```
7. To find files whose size is more than 10MB & less than 60MB in /root
        ~]#find . -size +10M  -size  -60M



8. To find all files having extension .txt in entire file system
        ~]#find / -type f -iname  "*.txt"

9. To find all files having extension .txt  and starting with a character in entire file system
        ~]#find / -type f -iname  "a*.txt"

10. To find all files whose permission is set as 777 in /root
        ~]#find / -type f -perm 0777 -print

=====================================================================
Locate command is used to search a file/folder in Linux

1. #locate passwd
2. #locate inittab
3. #locate -c passwd               #-c will show the count value for passwd file existence in system
4. #touch testfile.txt
    #locate testfile.txt           will not show any result, as locate command is unable to search files

    #updatedb
    #locate testfile.txt           now result will show file path where it exist
```

User & Group Management V.IMP

1. /etc/passwd list of users, their uid, gid, home folder name, shell name etc
2. /etc/group list of groups, their gid, members name
3. /etc/shadow contains users password (encrypted form)
4. /etc/gshadow contains users profile settings

```
to create users:
#useradd john
#useradd tom
#passwd john            to set the password of john user
#passwd tom         to set the password of tom user


#id         to see the uid and gid & group membership details of currently logged in user
#id john        to see id details of john user

to view last 2 lines of /etc/passwd, /etc/group & /etc/shadow
#tail -2 /etc/passwd
#tail -2 /etc/group
#tail -2 /etc/shadow

#ls  /home          to see the list of home folders of users


passwd          to change/reset the password of own/self
```

`

```
Create user Smith with shell name as /bin/tcsh with uid as 600 and Secondary group as red, blue, green and Primary as Colors, and default home folder as myhome.


groupadd red
groupadd green
groupadd blue
groupadd Colors
mkdir  /myhome

#useradd   -u  600  -g   Colors -G   red,blue,green  -s /bin/tcsh  -d /myhome/Smith    Smith

-g      Primary
-G      Secondary
-s      shell
-d      default home folder
#tail -1 /etc/passwd
#id Smith
```

usermod groupmod

```
to delete user:
=============
userdel     u3          incorrect way of deletion

deletes user
deletes group
deletes pwd
will keep home folder/mailbox


#userdel  -r  u3               to delete user u3 along with all the associated data(including home folder and associated configuration)
=============================================================================================
#usermod   -u  3000  u1                to change uid of u1
#id  u1

#groupmod -g 3000  u1                  to change gid of u1
#id  u1
=============================================================================
to rename group  hr as Finance
#groupmod  --help
#groupmod  -n Finance  hr
tail -5 /etc/group
```

```
To change the default home folder for user as /newhome:
=======================================================
-m -->move   -d---destination directory
#mkdir  /newhome
#usermod -m  -d  /newhome/u1     u1



ls /newhome     u1
ls /home
```

```
# Adding a user mary,tom,smith to a group red & green:

#groupadd red; groupadd green
#useradd smith
#passwd smith
#gpasswd -M mary,smith,tom   red   #-M  members
#gpasswd -M mary,smith,tom   green

tail -2 /etc/group
```

```
to remove a user from group
# gpasswd -d tom red
```

Sudouser

To make user admin
Visudo

Line no 99
Add
Harry ALL=(ALL) ALL

And switch to user harry and try to add user using sudo adduser m1

```
To give sudo rights to jane user for specific commands:
                        ping    useradd  passwd  fdisk
#visudo

esc:se    nu                to generate line number


goto line 99
jane    ALL = /bin/ping, /usr/sbin/useradd, /usr/sbin/passwd, /sbin/fdisk

esc:wq          save

#su -l jane
#sudo useradd m1
#sudo passwd m1
#sudo ping localhost
```

```
  ================================================================================
   Alternate way to make jane a sudo user(admin right):
  ================================================
  #usermod -aG wheel jane

  #visudo

  esc:se    nu                to generate line number


  goto line 105
  uncomment(remove #)         %wheel   ALL=(ALL)   ALL
  esc:wq          save
  #su -l jane
  #sudo groupadd m1
  #tail -2 /etc/group
```

Alias

```
alias       creating a shortform of commands


#alias          to see all the available aliases
To create a new alias (temporary alias):
#alias m='mkdir'
#alias c='clear'

#alias

To create a permanent alias:
#cd
~]#vim   /root/.bashrc

i  for insert



alias c='clear'
alias p='pwd'
alias m='mkdir'
esc:wq              to save & exit
```

File Permissions

```
   owner     group     others

   root       root     rest of the users
```

```
folder
    read           ls
    write          mkdir  rmdir
    execute        cd /test

file

    read           cat less
    write          create, delete modify
    execute        applicable for executable files
```

```
ways to work with permissions:
1. numeric:          numbers to change
2. symbolic:             characters to change permission

    read    4
    write   2
    exec    1
```

chown & chgroup

```bash
chown hemant myfolder
```

To change ownership recursively, including everything inside a directory:

```bash
chown -R hemant myfolder
```

The `chgrp` command changes the group ownership of a file or directory.

Syntax:

```bash
chgrp groupname filename
```

Example:

```bash
chgrp developers file.txt
```

```
    Symbolic way to change permission:
                    #chmod   o-rx  /reports


        current         root        root        rwxr-xr-x           reports
        required                ow-rx, gr----, others--  fc


                    chmod  u-w,g-rx,o+w         /reports
                    chmod u=rx,g=---,o=rwx      /reports


        required    ow gr others --->  fc
                chmod ugo=rwx  /reports
                chmod a=rwx /reports
```

Sticky Bit

```
1. sticky bit

    If we set the sticky bit on a directory, a file/folder under this directory can be deleted/renamed/moved only by either the owner of the
file/directory or the root user. Sticky bit will restrict rest of the users from renaming/deleting/moving existing files/folders, however they can
work on their own data.

To set sticky bit:
===============
        1.#chmod 1777  /reports

            or
        2.#chmod o+t  /reports
To verify:
==========  ls -ld /reports

        drwxrwxrwt      root       root           reports

To remove sticky bit:
==================
        1.#chmod 0777  /reports

            or
        2.#chmod o-t  /reports
```

```
2. SUID (Set user ID):  4       s           working on owner(u)

A file with SUID always executes as the user who owns the file, regardless of the user passing the command. Any user can use the command set with
SUID bit, even though user is not the owner of the command.

#ll  /usr/bin/passwd        SUID bit is set on passwd, because of this, all the users can execute this command, although root is owner.

        #rwsrwxrwx      root            root        /usr/bin/passwd

To set SUID bit:
===============
        1.#chmod 4777  /usr/bin/passwd

            or
        2.#chmod u+s  /usr/bin/passwd
To verify:
==========  #ll  /usr/bin/passwd

        drwsrwxrwx      root        root            /usr/bin/passwd

To remove SUID bit:
==================
        1.#chmod 0777  /usr/bin/passwd

            or
        2.#chmod u-s  /usr/bin/passwd
```

```
3. SGID  (Set Group ID):        2       s           working on groups(g)

If the SGID bit is set on any directory, all subdirectories and files created inside will get the same group ownership as the main directory, it
doesn't matter who is creating it


To set SGID bit:
===============
        1.#chmod 2777  /reports

            or
        2.#chmod g+s  /reports
To verify:
==========  #ls -ld  /reports

        drwxrwsrwx      root        root            reports

login as user1:
        $cd /reports

        $touch a.txt b.txt

        $ls -ld  /reports

        -rwxrwxr--  user1       root            a.txt
        -rwxrwxr--  user1       root            b.txt

as above, owner is user1, as he logged in and created files, but duo to SGID bit is set, group ownership is retained to root only.

To remove SUID bit:
==================
        1.#chmod g-s  reports
```

—

Tar

tar -cvzf /home.tar.gz /home
tar -tvf /home.tar.gz
tar -xvf /home.tar.gz

CPIO

To take backup
Cd to destination dir
ls | cpio ov > /backup/notes.cpio

To restore
Cpio iv < /backup/notes.cpio

LVM

```
[root@hemant ~]# pvcreate /dev/nvme0n{2p1,2p2,3p1}
  Physical volume "/dev/nvme0n2p1" successfully created.
  Physical volume "/dev/nvme0n2p2" successfully created.
  Physical volume "/dev/nvme0n3p1" successfully created.
```

```
[root@hemant ~]# vgcreate dept /dev/nvme0n{2p1,2p2,3p1}
  Volume group "dept" successfully created
[root@hemant ~]# lvcreate
dept  rhel
[root@hemant ~]# lvcreate
dept  rhel
[root@hemant ~]# lvcreate -L +15G dept -n sales
  Logical volume "sales" created.
```

```
[root@hemant ~]# mkfs.ext4 /dev/dept/sales
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 3932160 4k blocks and 983040 inodes
Filesystem UUID: 76176fe4-1b54-457f-b8ae-febc16b07873
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 160563
Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[root@hemant ~]# vi /etc/fstab
```

1. Correct command

Since the filesystem is ext4, use:

Bash

```
lvresize -r -L -1G /dev/dept/sales
```

Pvcreate /dev/sda1
Vgextend dept /dev/sda1

Job Scheduling

At Jobs

At 1044
at> mkdir hello

Cron Jobs

crontab -e
To edit cornjob file

crontab -l
To view the cronjob

crontab -r
To remove all the jobs

*	*	*	*	*
Min	hr	date	month	day

Nice

```
nice -n 6 gnome-terminal &

renice -n -20 -p 4000

top -p id
To check the nice value

To get the ID use
Pidof gedit

kill -9 id #force full kill
```

```
kill -15 id # normal kill /default

pkill processname
Kill processID
```

Suid, Sgid, sticky bit

Suid 4
Sgid 2
Sticky 1

Operation	Command	Meaning
Set SUID	`chmod u+s file`	Add SUID
Remove SUID	`chmod u-s file`	Remove SUID
Set SGID	`chmod g+s file`	Add SGID
Remove SGID	`chmod g-s file`	Remove SGID
Set both	`chmod u+s,g+s file`	Add SUID and SGID

ACL

Bash

```
setfacl -m u:user2:rw data.txt
```

Gives `user2` read and write permissions.

Bash

```
setfacl -m u:user2:rwx data.txt
```

B. Give permissions to a specific group

Bash

```
setfacl -m g:developers:rw data.txt
```

C. Give multiple users permissions

Bash

```
setfacl -m u:user2:r,u:user3:rw data.txt
```

D. Remove a user's ACL entry

Bash

```
setfacl -x u:user2 data.txt
```

SCP

Ssh should work and after that you can do

4. Step 3: Send the file from server to client

Run this command **on the server**:

```bash
scp /home/server1/data.txt client1@192.168.1.20:/home/client1/
```

Explanation:

- `/home/server1/data.txt` — source file on the server.
- `client1` — username on the client.
- `192.168.1.20` — client IP address.
- `/home/client1/` — destination directory on the client.

1. Syntax

```bash
scp -r source_directory username@IP_ADDRESS:/destination/path/
```

CUPS

```
  Main Configuration file:            /etc/cups/cupsd.conf
  To configure Printer:
  #system-config-printer
```

```
cancel      Cancels a print request
lp      Submits a print request
lpq     Shows the current print queue status
lpr     Submits a print request
lprm        Cancels print jobs that have been queued for printing
lpstat      Displays the status information for queues and requests

# lpr /root/install.log     #to print a file
# lp /root/anaconda-ks.cfg      #to print a file
# lpq                   #to view the print queue
# lprm 1                #to cancel job id 1
```

Chrony time

```
on S1 machine-NTP Server:
=======================
#date 040121002026.30
date mmddhourminyear.sec

#yum info chrony
#yum install chrony -y

#vim  /etc/chrony.conf

esc:se   nu         #to generate line number.


comment line num 3 4 5 6

uncomment line number 26 and add nw address with cidr value (Its necessary here in training to get the immediate eff
allow 172.16.0.0/16         #To allow Local Clients to Access the Server


uncomment line number 29    local stratum 10
esc:x           save
```

```
#systemctl stop ntpd        #optional-->Stop and disable ntpd if it is running, as both services cannot run simultan
#systemctl disable ntpd     #optional

#systemctl enable chronyd
#systemctl restart chronyd

#chronyc sources            #List the configured sources
#chronyc tracking           #Provides detailed synchronization status information
```

Client

```
#yum info chrony
#yum install chrony -y          optional

#vim  /etc/chrony.conf

esc:se   nu         to generate line number.


comment line num 3 4 5 6


goto last line

server 172.16.0.1

esc:x           save
```

```
To start and enable the chronyd service:

#systemctl stop ntpd
#systemctl disable ntpd

#systemctl enable chronyd
#systemctl start chronyd


#chronyc sources -v         #To check if the server is reachable and synchronizing
#chronyc tracking
#ntpdate -u 172.16.0.1      #To manually synchronize the date/time from server(10.10.10.1)
#timedatectl
```

Ssh

Both client and server

systemctl restart sshd
systemctl enable sshd

systemctl stop firewalld
setenforece 0

Telnet

ntsysv

```
On C1 (telnet Client 10.10.10.2):
================================
#yum install telnet* -y
#yum install xinetd* -y
#systemctl enable xinetd
#systemctl restart xinetd
#systemctl enable telnet.socket
#systemctl restart telnet.socket
#setenforce 0
#firewall-cmd  --add-port=23/tcp  --permanent
#firewall-cmd --reload
#setenforce 0


On telnet server:
#useradd user1
#passwd user1
    ***
    ***

On telnet-client:
#telnet 10.10.10.1

login:user1
password:***
```

VNC

```
On VNC Server:
==============
#yum install tigervnc*  -y

#useradd vncuser1
#passwd vncuser1
#su -l user1
$vncpasswd
    ******
    ******
$exit

#cp /usr/lib/systemd/system/vncserver@.service  /etc/systemd/system/vncserver@:1.service
```

```
#systemctl daemon-reload
#systemctl restart vncserver@:1.service
#systemctl enable vncserver@:1.service
#firewall-cmd --permanent  --add-service=vnc-server
#firewall-cmd --reload

or

#systemctl stop firewalld
#setenforce 0


On VNC Client:
==============
#yum install tigervnc*  -y
#setenforce 0
To connect to VNC Server
#vncviewer 10.10.10.1:1
            password ******
```

rsyslog

```
on S1 machine-log server: Centralized log server:
======================
#yum install rsyslog*   -y      optional


#vim  /etc/rsyslog.conf


esc:se   nu         to generate line number.

uncomment line number 13 14  17  18


goto last line

*.*  /var/log/mylog.log

esc:x        save
```

```
#touch /var/log/mylog.log
#systemctl status rsyslog.service
#systemctl restart rsyslog.service
systemctl enable rsyslog.service
#setenforce 0
For continuous logging updates (from all clients):
#tail -f  /var/log/mylog.log


on C1 machine-log Client(Linux):
=======================
#yum install rsyslog*   -y      optional


#vim  /etc/rsyslog.conf


esc:se   nu         to generate line number.

uncomment line number 13 14  17  18


goto last line

*.*  @10.10.10.1:514

esc:x              save


#systemctl status rsyslog.service
#systemctl restart rsyslog.service
#systemctl enable rsyslog.service

#firewall-cmd --zone=public --permanent --add-port=514/udp
#firewall-cmd --zone=public --permanent --add-port=514/tcp
#firewall-cmd --reload
#setenforce 0
```

```
Verification:
=============
#useradd u11
#groupadd sales1
#logger "hello, I am client"

on S1 machine-log server:
#cat /var/log/mylog.log          to verify log (will have user, gro
#tail -f  /var/log/mylog.log
```

SHELL

1. Case statements allow you to manage multiple possible outcomes based on the va
2. It simplifies complex conditions with multiple choices, making it more readabl
3. The Bash case statement is a form of the conditional if-elif-else statement.
4. Structured way to handle multiple choices or patterns for a given value.

```
#vim case-test.sh
#!/bin/bash

read -p  "Enter a number between 1 and 3:  "  n

case $n in
    1|one)
        echo "You selected One."
        ;;                  # ;;  break
    2|two)
        echo "You selected Two."
        ;;
    3|three)
        echo "You selected Three."
        ;;

     *)
        echo "Invalid selection. Please enter 1, 2, or 3."
        ;;
esac
```

```
#!/bin/bash
# Q1: Area of Square, Rectangle, Triangle and Circle

echo "===== AREA CALCULATOR ====="
echo "1. Square"
echo "2. Rectangle"
echo "3. Triangle"
echo "4. Circle"
read -p "Enter your choice (1-4): " choice

case $choice in
  1)
    read -p "Enter side of square: " side
    area=$(echo "$side * $side" | bc -l)
    echo "Area of Square = $area"
    ;;
  2)
```